EX NO - 3.1 CREATE 3D MODEL FROM ORTHOGRAPHIC PROJECTION

AIM:

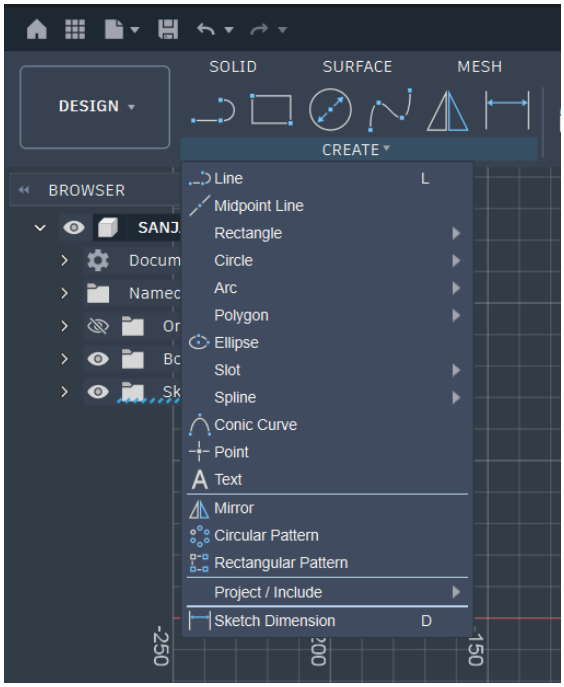
To sketch the given drawing with dimensions using fusion 360 tool.

REQUIREMENTS:

1. Autodesk fusion 360
2. Windows 11
3. 16 GB of RAM (integrated graphics recommend 2 GB or more)
4. 2.5 Mbps or faster download; 500 Kbps or faster upload

PROCEDURE:

1. Open a new file and then select the sketch icon.
2. Select the plane for sketching.
3. Use sketching features (line, circle, arc, rectangle, polygon, etc.,) to draw the given sketch.



1. Top View Dimensions

- **11 millimeters:** Thickness of the outer flange/fork walls and positioning distances.
- **18 millimeters:** Width of the inner slot opening (between the fork prongs).
- **52 millimeters:** Total overall length of the fork head component.

2. Front View Dimensions

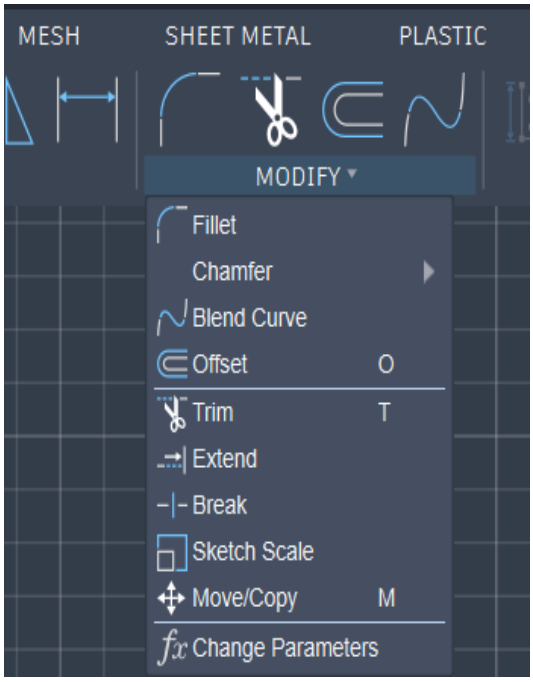
- **16 millimeters diameter:** Diameter of the circular pin hole ($\phi 16$).
- **16 millimeters radius:** Outer rounded radius of the fork end ($R16$).
- **36 millimeters:** Length of the main fork head body section.
- **32 millimeters:** Length of the cylindrical shaft/stem extending to the right.
- **18 millimeters diameter:** Diameter of the cylindrical shaft ($\phi 18$).

3. Right View Dimensions

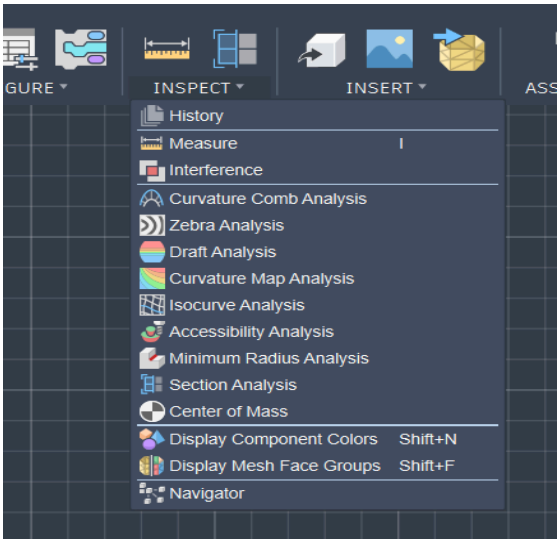
- **32 millimeters:** Height of the square/rectangular section of the right view.
- **40 millimeters:** Width/length of the right-hand block section

4. Use sketch modifying features (cut, trim, mirror, move, etc.,) to edit the sketch

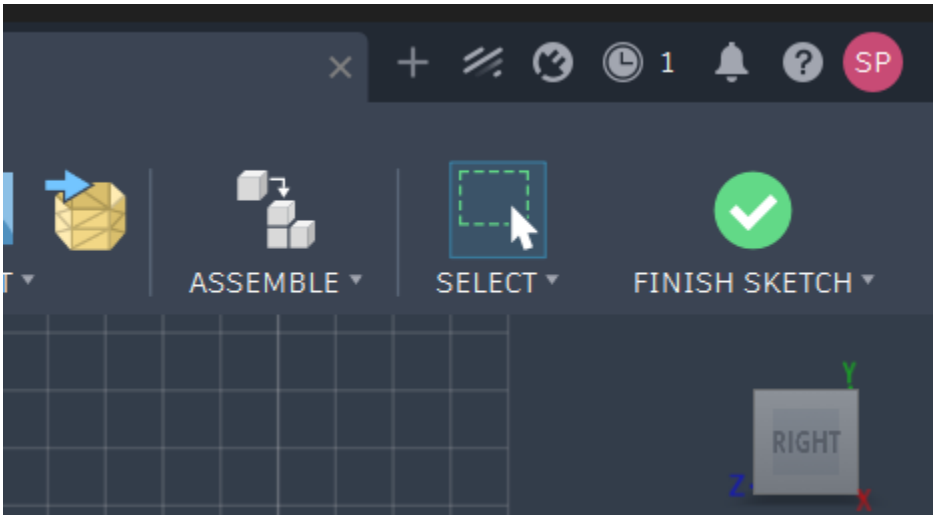
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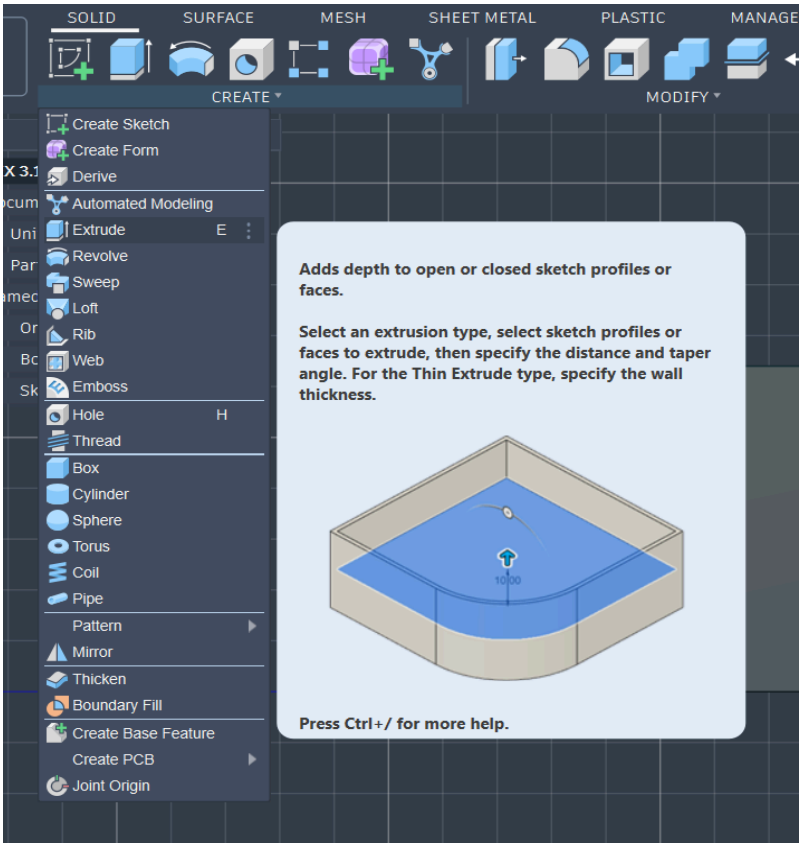
5. Use inspect feature to measure the dimensions of the sketch.



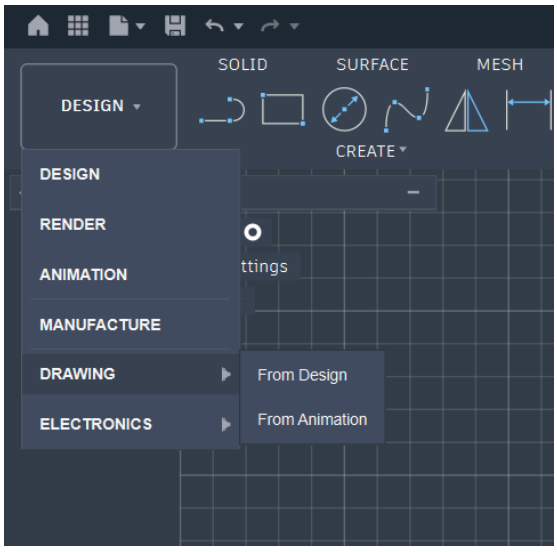
6. After finishing the sketch use the finish sketch option to finalize the sketch.



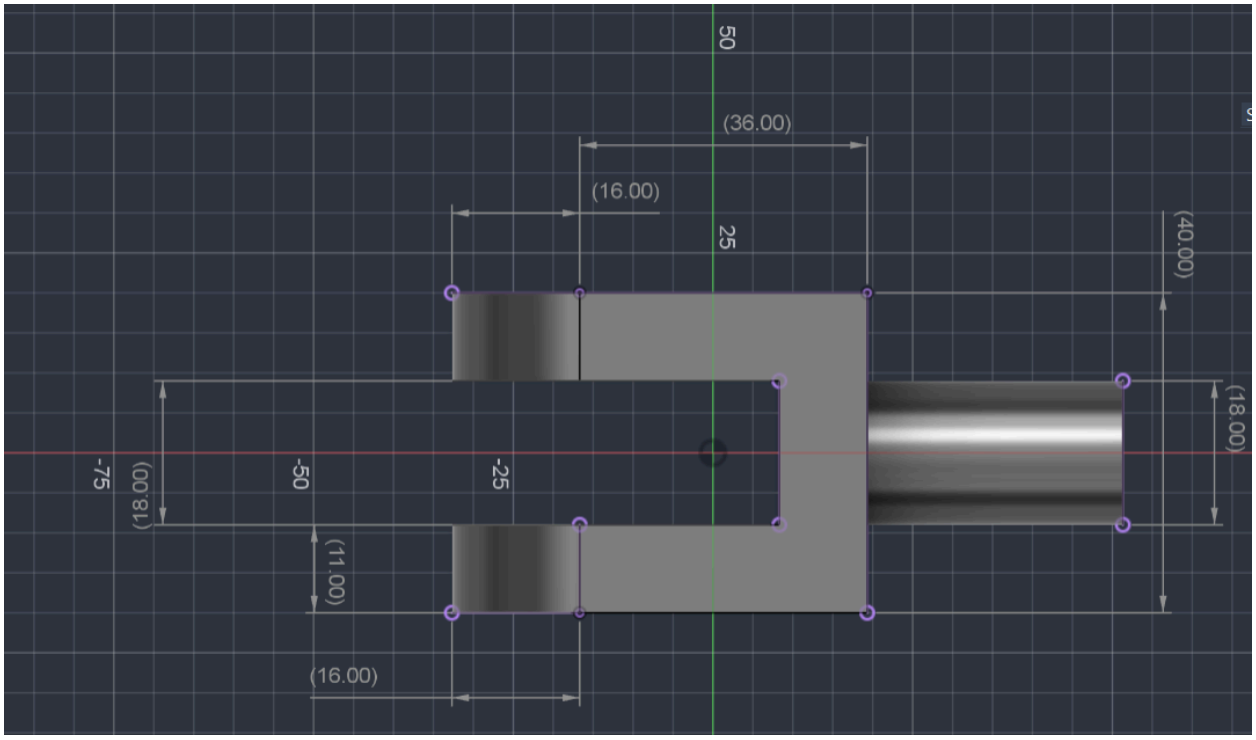
7. Use extrude option to convert 3D model from sketch



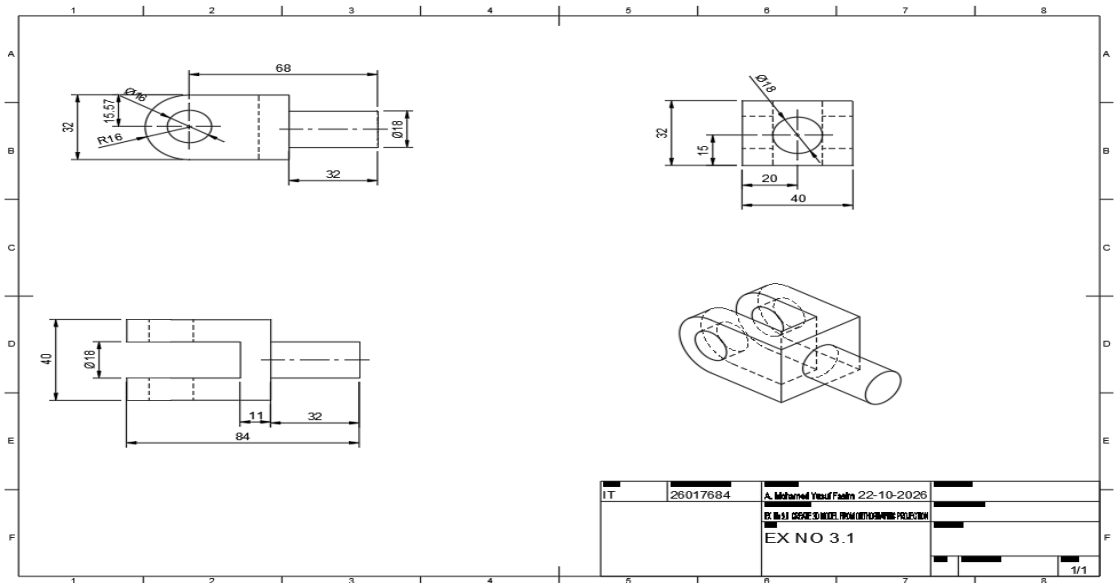
8. After finishing the model convert to drawing sheet



INPUT:



OUTPUT:



RESULT:

Thus, the 3D model is created using orthographic projection.